

■ EllipticTheta

`EllipticTheta[a, u, q]` gives the elliptic theta functions $\theta_a(u|q)$ ($a = 1, \dots, 4$).

Mathematical function (see Section ??). ■ $\theta_1(u, q) = 2q^{\frac{1}{4}} \sum_{n=0}^{\infty} (-1)^n q^{n(n+1)} \sin((2n+1)u)$,

$\theta_2(u, q) = 2q^{\frac{1}{4}} \sum_{n=0}^{\infty} q^{n(n+1)} \cos((2n+1)u)$, $\theta_3(u, q) = 1 + 2 \sum_{n=1}^{\infty} q^{n^2} \cos(2nu)$, $\theta_4(u, q) = 1 + 2 \sum_{n=1}^{\infty} (-1)^n q^{n^2} \cos(2nu)$.

■ See page 374.